

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used in data collection.
Data analysis	We used DeepSeek-R1 (DeepSeek), LLaMA 3.1 405B (Meta), and ChatGPT-o1(OpenAI)'s official scripts to evaluate the model performance on Azure Clouds. The Azure services used in this study are: https://ai.azure.com/explore/models/Meta-Llama-3.1-405B-Instruct/version/1/registry/azureml-meta?tid=77e89d61-570f-43b0-b9e4-634f462e34b8#details https://azuremarketplace.microsoft.com/en/marketplace/apps/metagenai.meta-llama-3-1-405b-instruct-offer?tab=overview https://ai.azure.com/explore/models/DeepSeek-R1/version/1/registry/azureml-deepseek?tid=77e89d61-570f-43b0-b9e4-634f462e34b8 SPSS v28.0 was used for statistical analysis.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The USMLE dataset is available at <https://www.usmle.org/exam-resources>. The NEJM Case Challenges are available at <https://www.nejm.org/case-challenges>. The BMJ Endgames are available at <https://www.bmj.com/specialties/endgames>. The MIMIC-III dataset can be requested at <https://physionet.org/content/mimiciii/1.4/>. The Mediciline dataset (<http://www.medicline.com/>) of multiple-choice questions for clinical diagnosis and management translated in English is available upon request for private/research use only, after agreement of Mediciline. Mediciline data requests should be sent to Mickael.Tordjman@mssm.edu. In response to the inquiry, the timeframe for responding to requests is approximately within 2 weeks of the request. The radiological reports used for summarization and RECIST classification are not available due to privacy issues.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

The NEJM and BMJ case challenges cover patients with diverse sex and gender. The imaging reports of the private dataset used for summarization and RECIST classification have a balanced sex distribution.

Reporting on race, ethnicity, or other socially relevant groupings

Race, ethnicity, or other socially relevant groupings were not relevant to the objectives or analyses of this study and therefore were not collected or analyzed. This study focused solely on the comparative performance of the Large Language Models across different medical tasks without considering these social variables.

Population characteristics

The population characteristics in this study are related to the challenges presented in the New England Journal of Medicine and the BMJ Endgames. These sources provided the basis for the tasks assigned to the evaluated Large Language Models, focusing on their application in diverse and complex analytical settings.

Recruitment

The reports of the private dataset were randomly selected from the Radiology Imaging System between 2010 and 2023 for the summarization task. Two cancer imaging reports of chest-abdomen-pelvis CT-scans were selected consecutively for the RECIST classification task. For the case challenges of the medical journals, we included all available cases from NEJM case challenges and the most recent cases from BMJ Endgames including a management section.

Ethics oversight

This study was approved by the Institutional Review Board of the Icahn School of Medicine at Mount Sinai to use the anonymized diagnostic imaging reports, in accordance with the institution's Federalwide Assurances (FWA00005656, FWA00005651), under IRB protocol number GCO# 20-2199-00001-01.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

The publicly available datasets used in this study include NEJM Case Challenges published up to March 5, 2025, the 50 most recent BMJ Endgames management cases up to the same date, and the full set of USMLE questions (excluding image-based cases), all sourced from online resources. For the MIMIC-III dataset, we randomly sampled 20 cases per imaging modality (CT, MRI, PET, X-ray, and ultrasound) to ensure balanced coverage across modalities while maintaining feasibility for manual evaluation and computational analysis. For report summarization, we also randomly sampled 20 cases per imaging modality (CT, MRI, PET, X-ray, and ultrasound) from an additional private dataset. For RECIST 1.1 classification, two consecutive cancer imaging reports of 100 patients were randomly selected from clinical notes. 200 questions were randomly chosen for the Mediciline QA task. These sample sizes were chosen to provide adequate representation for each task while ensuring that performance evaluation could be conducted effectively within the scope of this study.

Data exclusions

This study strictly focuses on text-based evaluations of LLM capabilities. Image-based cases from NEJM, BMJ, and USMLE were excluded during the study design because they require multi-modal version of LLMs.

Replication	Each model was evaluated three times per task on Azure Cloud, with all runs completing successfully.
Randomization	The imaging reports were randomly selected from the RIS system between 2010 and 2023. Two cancer imaging reports of chest-abdomen-pelvis CT-scans were selected consecutively for the RECIST classification task. The “Findings” section of 200 imaging reports, 100 from an outpatient radiology facility in the New York Metropolitan area and 100 from MIMIC reports (each with 20 examples of X-rays, ultrasound, CT-scan, MRI and PET/CT reports), were provided to the three LLMs for summarization. Since our study evaluates the performance of LLMs in summarizing medical reports rather than assessing clinical interventions or comparing patient outcomes, random allocation or covariate control typically applied in interventional studies is not relevant here.
Blinding	The human readers participating in evaluation of the report summarization generated by the LLMs were blinded from which model provided the outputs.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.